Uncertainty-Aware Multi-Agent Ransomware Detection via Decision MLP Fusion, Score-Space Augmentation, and Bayesian Triage Policy Optimisation

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Why Existing Detectors Fall Short

- **No principled uncertainty** — a deterministic softmax score cannot distinguish a confident-correct prediction from a confident-wrong one.
- **Heuristic modality fusion** — equal-weight averaging of static PE features and runtime telemetry discards per-modality reliability.
- **Binary triage** — alerting on every uncertain case overwhelms analysts; auto-containing every uncertain case risks business disruption.

Four Contributions

1. A **heterogeneous three-agent architecture** (Semantic, Behavioural, Knowledge CNNs), each producing a probability *and* an epistemic uncertainty via T=10 MC Dropout passes.
2. A **Decision MLP fusion** of the six-dimensional score pool, temperature-calibrated (ECE 0.027), with **CTGAN** score-space augmentation for class imbalance.
3. A **Bayesian TPE triage policy** over four jointly-optimised thresholds, cutting escalation from 58.3%→30.6% while raising balanced accuracy 0.753→0.864.
4. **Peer Consultation** against knowledge collapse under concurrent execution, and an **Expert Feedback Loop** (EWC) for continual adaptation with negligible forgetting (ΔAUPRC=0.001).

Datasets

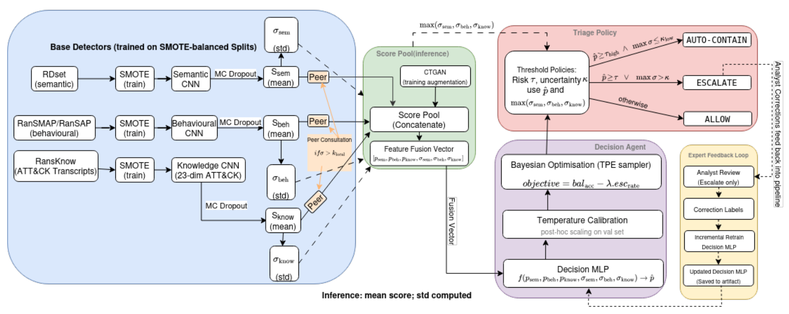
- **RDset**: 138,047 PE files (96,724 ransomware / 41,323 benign) — Semantic Agent.
- **RanSMAP**: 35,857 ten-second hypervisor telemetry windows, 1,776 sessions, 15 hardware configurations, six ransomware families — Behavioural Agent.
- **RansKnow** (ours): 718 ATT&CK-annotated threat-intelligence transcripts — Knowledge Agent.

RDset, RanSMAP, and RansKnow are independently collected: no executable appears in more than one dataset, so Semantic and Knowledge contribute a fixed, label-independent corpus-mean score during RanSMAP fusion — only the Behavioural Agent carries per-sample signal there.

Limitations & Future Work

- Concurrent benign+ransomware execution remains hard for the most adversarial pairing (SDelete+REvil); Peer Consultation partially recovers detection but the system still escalates rather than confidently contains.
- RanSMAP covers families through 2022; extension to Cl0p, BlackCat/ALPHV, LockBit 3.0 is future work.
- An adaptive attacker could in principle suppress $\sigma_{\max}$ while keeping $\hat{p}$ sub-threshold; online Bayesian recalibration of triage thresholds under concept drift is likewise deferred to future work.

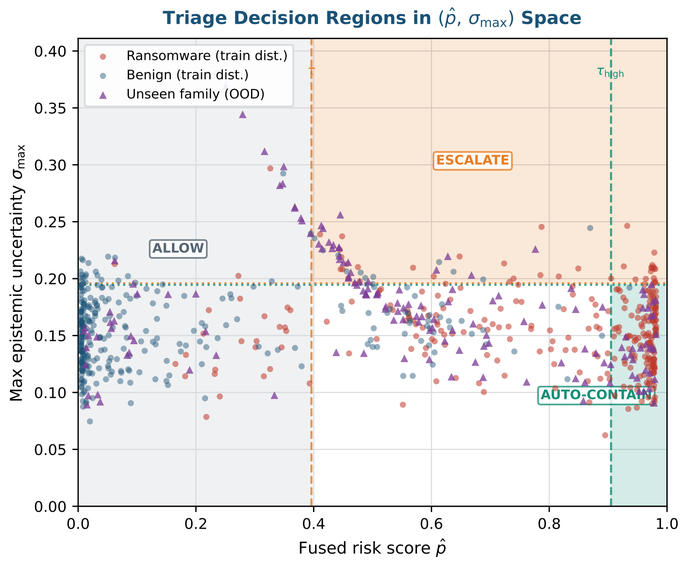
Framework Overview



Three specialised CNN agents each produce $(\hat{p}_i, \sigma_i)$ via MC Dropout. Peer Consultation substitutes an unreliable agent's probability before the score pool is fused by the Decision MLP into a calibrated risk $\hat{p}$. Bayesian TPE derives the four-threshold policy mapping $(\hat{p}, \sigma_{\max})$ to Auto-Contain, Escalate, or Allow; analyst corrections feed back via EWC-regularised incremental retraining.

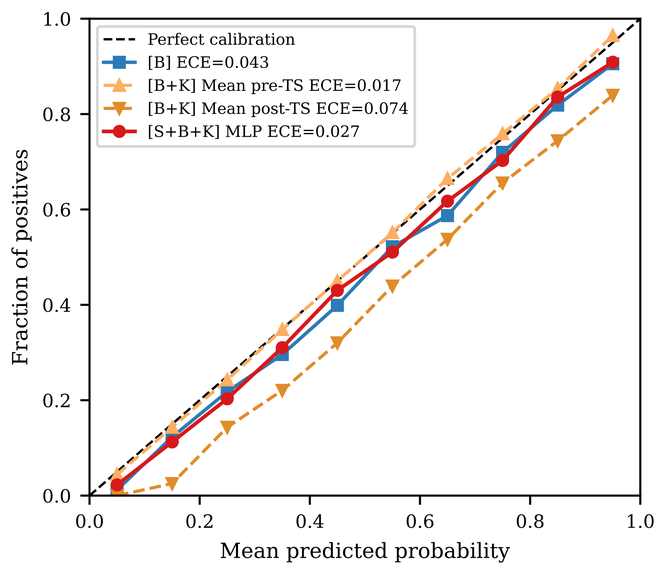
Triage Policy over $(\hat{p}, \sigma_{\max})$ Space

$$\pi(\hat{p}, \sigma_{\max}; \Theta) = \begin{cases} \textsc{Auto-Contain} & \hat{p} \geq \tau_{\text{high}} \wedge \sigma_{\max} \leq \kappa_{\text{low}} \\ \textsc{Escalate} & \hat{p} \geq \tau \vee \sigma_{\max} > \kappa \\ \textsc{Allow} & \text{otherwise} \end{cases}$$



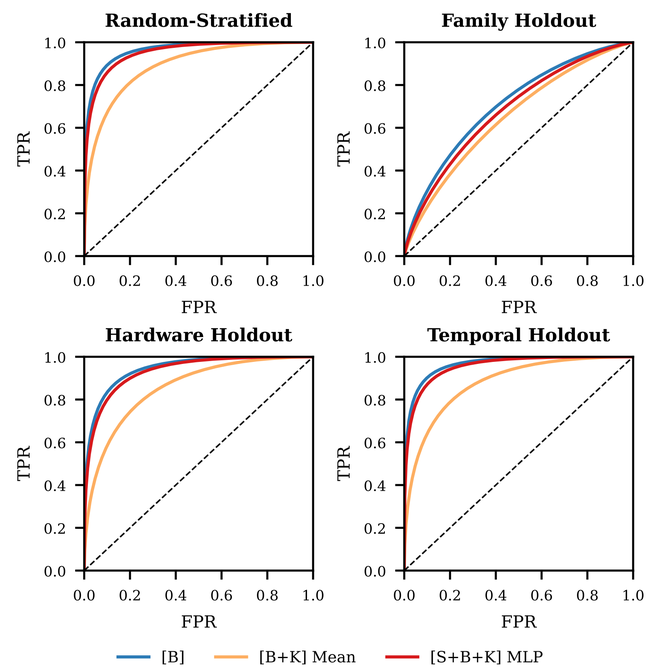
Out-of-distribution samples from unseen families concentrate in the Escalate region (63%) due to elevated $\sigma_{\max}$, converting would-be silent misses into analyst review.

Calibration: Fusion is a Calibration Story, not an AUROC Story



Since Semantic/Knowledge contribute fixed corpus-mean scores on RanSMAP, the Decision MLP cannot exceed the Behavioural Agent's own AUROC here. Its honest value is uncertainty-gated triage: naive mean fusion is in fact better calibrated (ECE 0.017) than the Decision MLP (ECE 0.027).

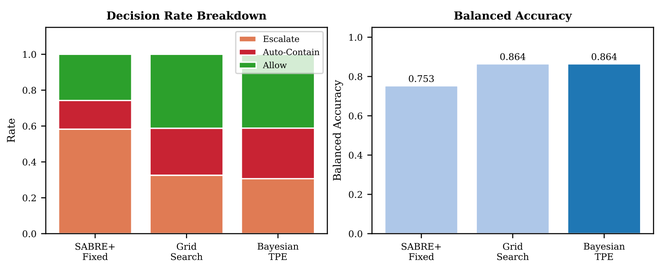
Detection Performance



System	Random	Temporal	Family	Hardware
Semantic (RDset)	**0.996**	–	–	–
Behavioural	**0.961**	**0.960**	**0.707**	**0.939**
Decision MLP	0.952	0.956	0.681	0.933

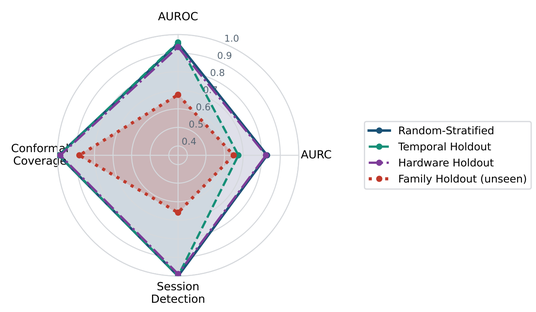
Family holdout is the one generalisation gap: AUROC falls to 0.707, but elevated σ_{beh} correctly routes most novel-family samples to escalation rather than silent miss.

Triage Policy Optimisation



Policy	Bal. Acc. ↑	Escalate (%) ↓	Auto-C (%) ↑
Fixed thresholds	0.753	58.3	16.0
Grid search	0.864	32.6	26.2
Bayesian TPE	**0.864**	**30.6**	**28.2**

Generalisation Across Four Splits



Random-stratified, temporal, and hardware holdout cluster tightly on every axis (AUROC, selective precision-coverage, session-level detection, Mondrian conformal coverage); family holdout collapses inward on all four — the same generalisation gap, not four independent weaknesses.

Conclusion

A three-agent framework that treats epistemic uncertainty as a **first-class control variable** for autonomous triage: strong in-distribution detection (AUROC 0.996 static / 0.963 behavioural), a 47% reduction in unnecessary escalation, and a Mondrian conformal wrapper giving distribution-free coverage guarantees where exchangeability holds.

`github.com/Henrinnes/Agentic_SABRE`

LaTeX TikZposter